

Flag 1 – Prepare Your Environment

Objective: Install Node.js.

Screenshot:

- The command/The installer required for installing Node.js
- Run :
node -v
npm -v

```
C:\Users\Student>node -v  
v26.5.0
```

Flag 2 — Unlock the Forge

Objective: Install Angular CLI.

Screenshot:

- The command to install the CLI
- Then run:
ng version

```
C:\Users\Student>ng version  
  
Angular CLI  
Node.js  
Package Manager  
Operating System  
: 22.0.8  
: 26.5.0  
: npm 12.0.1  
: win32 x64
```

Flag 3 — Forge Your Project

Objective: Create your first Angular project.

Screenshot:

- The command to create the project

```
C:\Users\Student>ng new capture-the-flag --routing --style=scss
✓ Do you want to enable Server-Side Rendering (SSR) and Static Site Generation (SSG/Prerendering)? Yes
✓ Which AI tools do you want to configure with Angular best practices? https://angular.dev/ai/develop-with-ai None
CREATE capture-the-flag/.prettierrc (173 bytes)
CREATE capture-the-flag/angular.json (2378 bytes)
CREATE capture-the-flag/package.json (1036 bytes)
CREATE capture-the-flag/README.md (1526 bytes)
CREATE capture-the-flag/tsconfig.json (939 bytes)
CREATE capture-the-flag/.editorconfig (331 bytes)
CREATE capture-the-flag/.gitignore (666 bytes)
CREATE capture-the-flag/tsconfig.app.json (432 bytes)
CREATE capture-the-flag/tsconfig.spec.json (456 bytes)
CREATE capture-the-flag/.vscode/extensions.json (134 bytes)
CREATE capture-the-flag/.vscode/launch.json (490 bytes)
CREATE capture-the-flag/.vscode/tasks.json (1020 bytes)
CREATE capture-the-flag/src/main.ts (228 bytes)
CREATE capture-the-flag/src/index.html (313 bytes)
CREATE capture-the-flag/src/styles.scss (81 bytes)
CREATE capture-the-flag/src/main.server.ts (300 bytes)
CREATE capture-the-flag/src/server.ts (1677 bytes)
CREATE capture-the-flag/src/app/app.spec.ts (706 bytes)
CREATE capture-the-flag/src/app/app.ts (311 bytes)
CREATE capture-the-flag/src/app/app.scss (0 bytes)
CREATE capture-the-flag/src/app/app.html (20488 bytes)
CREATE capture-the-flag/src/app/app.config.ts (419 bytes)
CREATE capture-the-flag/src/app/app.routes.ts (80 bytes)
CREATE capture-the-flag/src/app/app.config.server.ts (438 bytes)
CREATE capture-the-flag/src/app/app.routes.server.ts (174 bytes)
CREATE capture-the-flag/public/favicon.ico (15086 bytes)
✓ Packages installed successfully.
warning: in the working copy of 'package-lock.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'package.json', LF will be replaced by CRLF the next time Git touches it
Successfully initialized git.
```

```
C:\Users\Student>cd capture-the-flag
```

```
C:\Users\Student\capture-the-flag>ng generate component pages/login
```

```
CREATE src/app/pages/login/login.spec.ts (547 bytes)
```

```
CREATE src/app/pages/login/login.ts (194 bytes)
```

```
CREATE src/app/pages/login/login.scss (0 bytes)
```

```
CREATE src/app/pages/login/login.html (21 bytes)
```

```
C:\Users\Student\capture-the-flag>ng generate component pages/dashboard
```

```
CREATE src/app/pages/dashboard/dashboard.spec.ts (575 bytes)
```

```
CREATE src/app/pages/dashboard/dashboard.ts (210 bytes)
```

```
CREATE src/app/pages/dashboard/dashboard.scss (0 bytes)
```

```
CREATE src/app/pages/dashboard/dashboard.html (25 bytes)
```

```
C:\Users\Student\capture-the-flag>ng generate component pages/challenges
```

```
CREATE src/app/pages/challenges/challenges.spec.ts (582 bytes)
```

```
CREATE src/app/pages/challenges/challenges.ts (214 bytes)
```

```
CREATE src/app/pages/challenges/challenges.scss (0 bytes)
```

```
CREATE src/app/pages/challenges/challenges.html (26 bytes)
```

```
C:\Users\Student\capture-the-flag>ng generate component pages/challenge-details
```

```
CREATE src/app/pages/challenge-details/challenge-details.spec.ts (625 bytes)
```

```
CREATE src/app/pages/challenge-details/challenge-details.ts (241 bytes)
```

```
CREATE src/app/pages/challenge-details/challenge-details.scss (0 bytes)
```

```
CREATE src/app/pages/challenge-details/challenge-details.html (33 bytes)
```

```
C:\Users\Student\capture-the-flag>ng generate component pages/leaderboard
```

```
CREATE src/app/pages/leaderboard/leaderboard.spec.ts (589 bytes)
```

```
CREATE src/app/pages/leaderboard/leaderboard.ts (218 bytes)
```

```
CREATE src/app/pages/leaderboard/leaderboard.scss (0 bytes)
```

```
CREATE src/app/pages/leaderboard/leaderboard.html (27 bytes)
```

```
C:\Users\Student\capture-the-flag>ng generate component pages/profile
```

```
CREATE src/app/pages/profile/profile.spec.ts (561 bytes)
```

```
CREATE src/app/pages/profile/profile.ts (202 bytes)
```

```
CREATE src/app/pages/profile/profile.scss (0 bytes)
```

```
CREATE src/app/pages/profile/profile.html (23 bytes)
```

```
C:\Users\Student\capture-the-flag>ng generate component shared/navbar
```

```
CREATE src/app/shared/navbar/navbar.spec.ts (554 bytes)
```

```
CREATE src/app/shared/navbar/navbar.ts (198 bytes)
```

```
CREATE src/app/shared/navbar/navbar.scss (0 bytes)
```

```
CREATE src/app/shared/navbar/navbar.html (22 bytes)
```

```
C:\Users\Student\capture-the-flag>ng generate component shared/footer
```

```
CREATE src/app/shared/footer/footer.spec.ts (554 bytes)
```

```
CREATE src/app/shared/footer/footer.ts (198 bytes)
```

```
CREATE src/app/shared/footer/footer.scss (0 bytes)
```

```
CREATE src/app/shared/footer/footer.html (22 bytes)
```

```
C:\Users\Student\capture-the-flag>ng g s services/auth
CREATE src/app/services/auth.spec.ts (327 bytes)
CREATE src/app/services/auth.ts (84 bytes)

C:\Users\Student\capture-the-flag>ng g s services/challenge
CREATE src/app/services/challenge.spec.ts (352 bytes)
CREATE src/app/services/challenge.ts (89 bytes)

C:\Users\Student\capture-the-flag>ng g s services/leaderboard
CREATE src/app/services/leaderboard.spec.ts (362 bytes)
CREATE src/app/services/leaderboard.ts (91 bytes)

C:\Users\Student\capture-the-flag>ng g s services/user
CREATE src/app/services/user.spec.ts (327 bytes)
CREATE src/app/services/user.ts (84 bytes)

C:\Users\Student\capture-the-flag>ng g s services/api
CREATE src/app/services/api.spec.ts (322 bytes)
CREATE src/app/services/api.ts (83 bytes)

C:\Users\Student\capture-the-flag>ng g interface models/user
CREATE src/app/models/user.ts (28 bytes)

C:\Users\Student\capture-the-flag>ng g interface models/challenge
CREATE src/app/models/challenge.ts (33 bytes)

C:\Users\Student\capture-the-flag>ng g interface models/flag
CREATE src/app/models/flag.ts (28 bytes)

C:\Users\Student\capture-the-flag>ng g interface models/leaderboard
CREATE src/app/models/leaderboard.ts (35 bytes)

C:\Users\Student\capture-the-flag>ng g guard guards/auth
✓ Which type of guard would you like to create? CanActivate
CREATE src/app/guards/auth-guard.spec.ts (478 bytes)
CREATE src/app/guards/auth-guard.ts (133 bytes)

C:\Users\Student\capture-the-flag>ng g guard guards/admin
✓ Which type of guard would you like to create? CanActivate
CREATE src/app/guards/admin-guard.spec.ts (482 bytes)
CREATE src/app/guards/admin-guard.ts (134 bytes)

C:\Users\Student\capture-the-flag>ng g pipe pipes/difficulty
CREATE src/app/pipes/difficulty-pipe.spec.ts (211 bytes)
CREATE src/app/pipes/difficulty-pipe.ts (238 bytes)

C:\Users\Student\capture-the-flag>ng g directive directives/highlight
CREATE src/app/directives/highlight.spec.ts (207 bytes)
CREATE src/app/directives/highlight.ts (124 bytes)

C:\Users\Student\capture-the-flag>ng g interceptor interceptors/auth
CREATE src/app/interceptors/auth-interceptor.spec.ts (493 bytes)
CREATE src/app/interceptors/auth-interceptor.ts (154 bytes)
```

```

C:\Users\Student\capture-the-flag>ng build

(node:4052) [DEP0205] DeprecationWarning: `module.register()` is deprecated. Use `module.registerHooks()` instead.
(Use `node --trace-deprecation ...` to show where the warning was created)
(node:4052) [DEP0205] DeprecationWarning: `module.register()` is deprecated. Use `module.registerHooks()` instead.
(Use `node --trace-deprecation ...` to show where the warning was created)
Browser bundles
Initial chunk files | Names          | Raw size | Estimated transfer size
main-EQRD44RR.js   | main           | 253.21 kB | 70.68 kB
styles-5INURTS0.css | styles         | 0 bytes  | 0 bytes

| Initial total   | 253.21 kB | 70.68 kB

Server bundles
Initial chunk files | Names          | Raw size
server.mjs          | server         | 816.76 kB
main.server.mjs     | main.server    | 691.77 kB
polyfills.server.mjs | polyfills.server | 234.18 kB
chunk-IT0R7N0UX.mjs | -              | 1.38 kB
chunk-KYQ0YI4M.mjs | -              | 1.01 kB

Lazy chunk files   | Names          | Raw size
chunk-KUERU4XA.mjs | xhr2           | 12.17 kB
chunk-LYHAGGDF.mjs | xhr2           | 12.07 kB

Prerendered 1 static route.
Application bundle generation complete. [29.684 seconds] - 2026-07-29T02:08:52.345Z

Output location: C:\Users\Student\capture-the-flag\dist\capture-the-flag

C:\Users\Student\capture-the-flag>ng build --configuration production

(node:20772) [DEP0205] DeprecationWarning: `module.register()` is deprecated. Use `module.registerHooks()` instead.
(Use `node --trace-deprecation ...` to show where the warning was created)
(node:20772) [DEP0205] DeprecationWarning: `module.register()` is deprecated. Use `module.registerHooks()` instead.
(Use `node --trace-deprecation ...` to show where the warning was created)
Browser bundles
Initial chunk files | Names          | Raw size | Estimated transfer size
main-EQRD44RR.js   | main           | 253.21 kB | 70.68 kB
styles-5INURTS0.css | styles         | 0 bytes  | 0 bytes

| Initial total   | 253.21 kB | 70.68 kB

Server bundles
Initial chunk files | Names          | Raw size
server.mjs          | server         | 816.76 kB
main.server.mjs     | main.server    | 691.77 kB
polyfills.server.mjs | polyfills.server | 234.18 kB
chunk-IT0R7N0UX.mjs | -              | 1.38 kB
chunk-KYQ0YI4M.mjs | -              | 1.01 kB

Lazy chunk files   | Names          | Raw size
chunk-KUERU4XA.mjs | xhr2           | 12.17 kB
chunk-LYHAGGDF.mjs | xhr2           | 12.07 kB

Prerendered 1 static route.
Application bundle generation complete. [3.543 seconds] - 2026-07-29T02:10:15.946Z

Output location: C:\Users\Student\capture-the-flag\dist\capture-the-flag

```

```

C:\Users\Student\capture-the-flag>ng test
Application bundle generation complete. [1.471 seconds] - 2026-07-29T02:10:28.664Z

Watch mode enabled. Watching for file changes...

DEV v4.1.10 C:/Users/Student/capture-the-flag

capture-the-flag src/app/pages/challenges/challenges.spec.ts (1 test) 47ms
capture-the-flag src/app/pages/dashboard/dashboard.spec.ts (1 test) 47ms
capture-the-flag src/app/pages/challenge-details/challenge-details.spec.ts (1 test) 48ms
capture-the-flag src/app/pages/leaderboard/leaderboard.spec.ts (1 test) 49ms
capture-the-flag src/app/pages/profile/profile.spec.ts (1 test) 53ms
capture-the-flag src/app/guards/admin-guard.spec.ts (1 test) 1ms
capture-the-flag src/app/interceptors/auth-interceptor.spec.ts (1 test) 1ms
capture-the-flag src/app/shared/footer/footer.spec.ts (1 test) 13ms
capture-the-flag src/app/shared/navbar/navbar.spec.ts (1 test) 18ms
capture-the-flag src/app/pages/login/login.spec.ts (1 test) 13ms
capture-the-flag src/app/guards/auth-guard.spec.ts (1 test) 1ms
capture-the-flag src/app/services/leaderboard.spec.ts (1 test) 2ms
capture-the-flag src/app/services/challenge.spec.ts (1 test) 3ms
capture-the-flag src/app/services/auth.spec.ts (1 test) 3ms
capture-the-flag src/app/pipes/difficulty-pipe.spec.ts (1 test) 1ms
capture-the-flag src/app/services/user.spec.ts (1 test) 4ms
capture-the-flag src/app/services/api.spec.ts (1 test) 4ms
capture-the-flag src/app/directives/highlight.spec.ts (1 test) 0ms
capture-the-flag src/app/app.spec.ts (2 tests) 91ms

Test Files 19 passed (19)
Tests 20 passed (20)
Start at 10:10:29
Duration 2.92s (transform 468ms, setup 2.90s, import 425ms, tests 399ms, environment 37.61s)

PASS Waiting for file changes...
press h to show help, press q to quit

C:\Users\Student\capture-the-flag>ng e2e
Cannot find "e2e" target for the specified project.
You can add a package that implements these capabilities.

For example:
Playwright: ng add playwright-ng-schematics
Cypress: ng add @cypress/schematic
Nightwatch: ng add @nightwatch/schematics
WebdriverIO: ng add @wdio/schematics
Puppeteer: ng add @puppeteer/ng-schematics

```

```
C:\Users\Student\capture-the-flag>ng e2e
Cannot find "e2e" target for the specified project.
You can add a package that implements these capabilities.

For example:
  Playwright: ng add playwright-ng-schematics
  Cypress: ng add @cypress/schematic
  Nightwatch: ng add @nightwatch/schematics
  WebdriverIO: ng add @wdio/schematics
  Puppeteer: ng add @puppeteer/ng-schematics

Would you like to add a package with "e2e" capabilities now? Puppeteer
✓ Determining Package Manager
  > Using package manager: npm
✓ Searching for compatible package version
  > Found compatible package version: 0.8.0.
✓ Loading package information
✗ Command aborted
■ Installing package
Command aborted

C:\Users\Student\capture-the-flag>ng lint
Cannot find "lint" target for the specified project.
You can add a package that implements these capabilities.

For example:
  ESLint: ng add angular-eslint

Would you like to add ESLint now? No

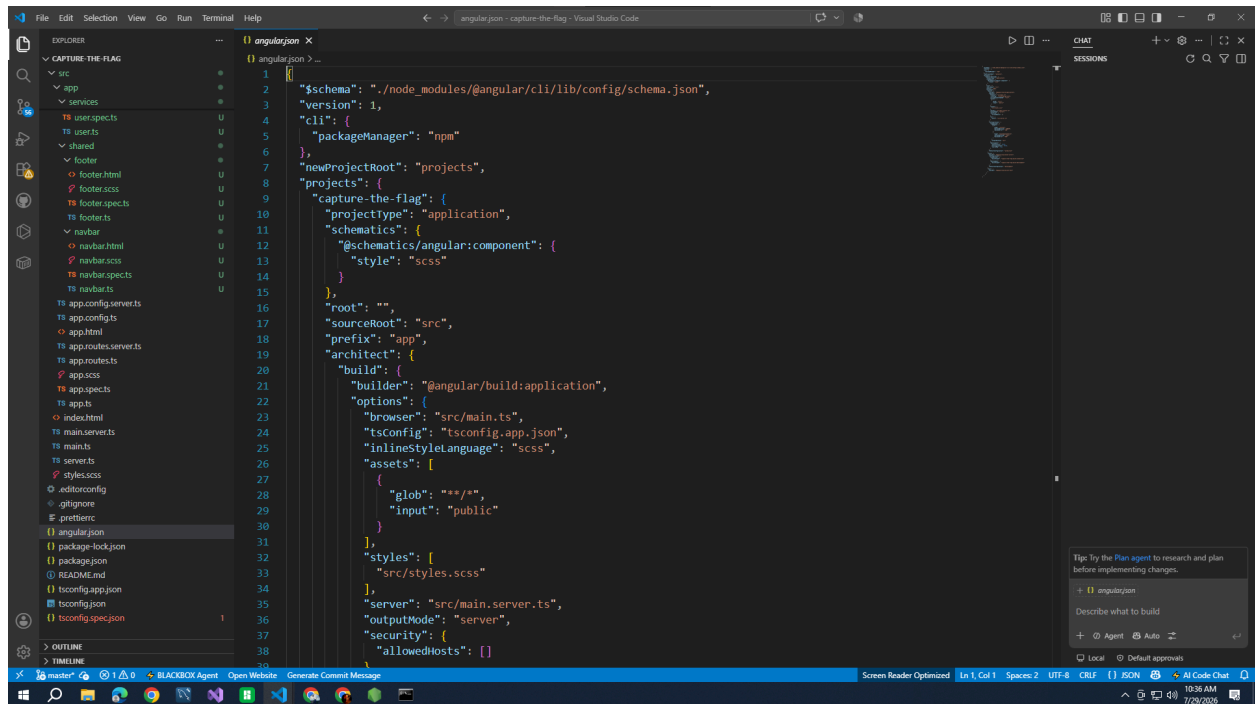
C:\Users\Student\capture-the-flag>
```

Flag 4 — Power Up

Objective: Open the project in VS Code.

Screenshot:

- Your VS Code containing the new project



Flag 5 — Launch

Objective: Run the development server.

Screenshot:

- The command to run the development server
- The success message

```
Server bundles
Initial chunk files | Names | Raw size
main.server.mjs | main.server | 49.86 kB
server.mjs | server | 1.84 kB
polyfills.server.mjs | polyfills.server | 268 bytes

Application bundle generation complete. [2.237 seconds] - 2026-07-29T02:45:22.765Z

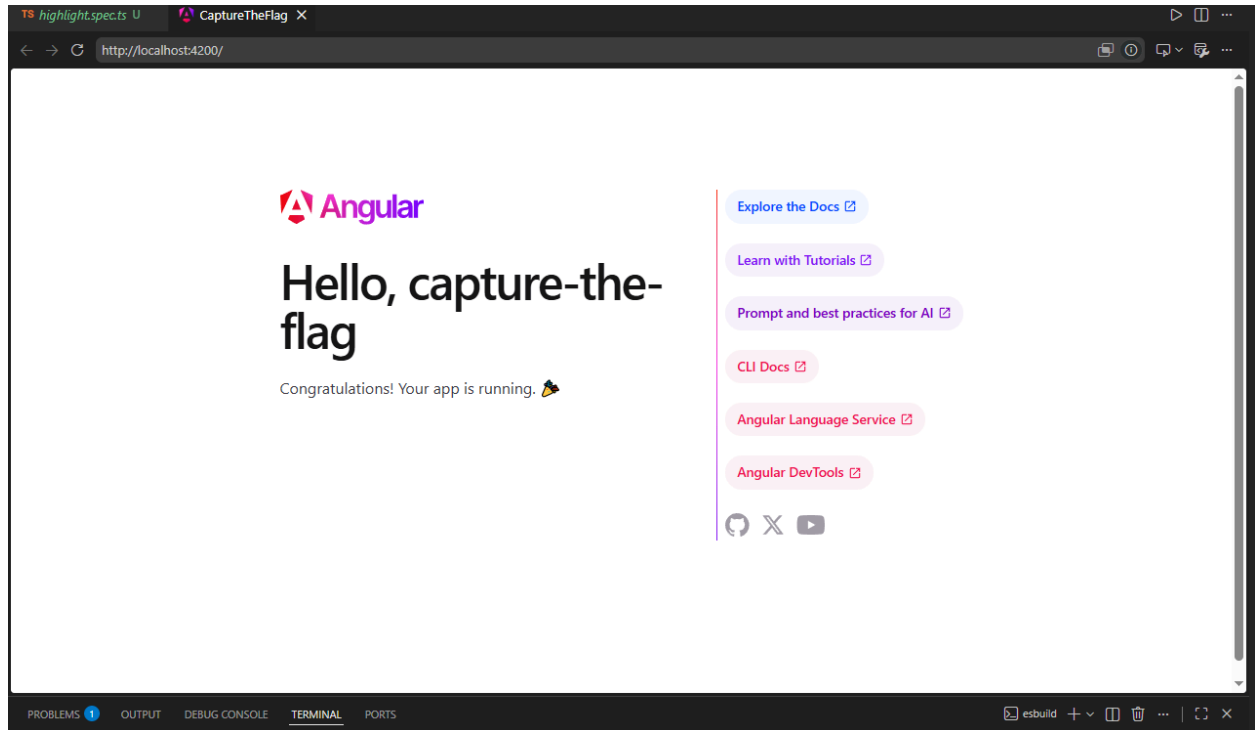
Watch mode enabled. Watching for file changes...
NOTE: Raw file sizes do not reflect development server per-request transformations.
→ Local: http://localhost:4200/
→ press h + enter to show help
```

Flag 6 — I See It

Objective: Open the application.

Screenshot:

- The browser
- The port it opened

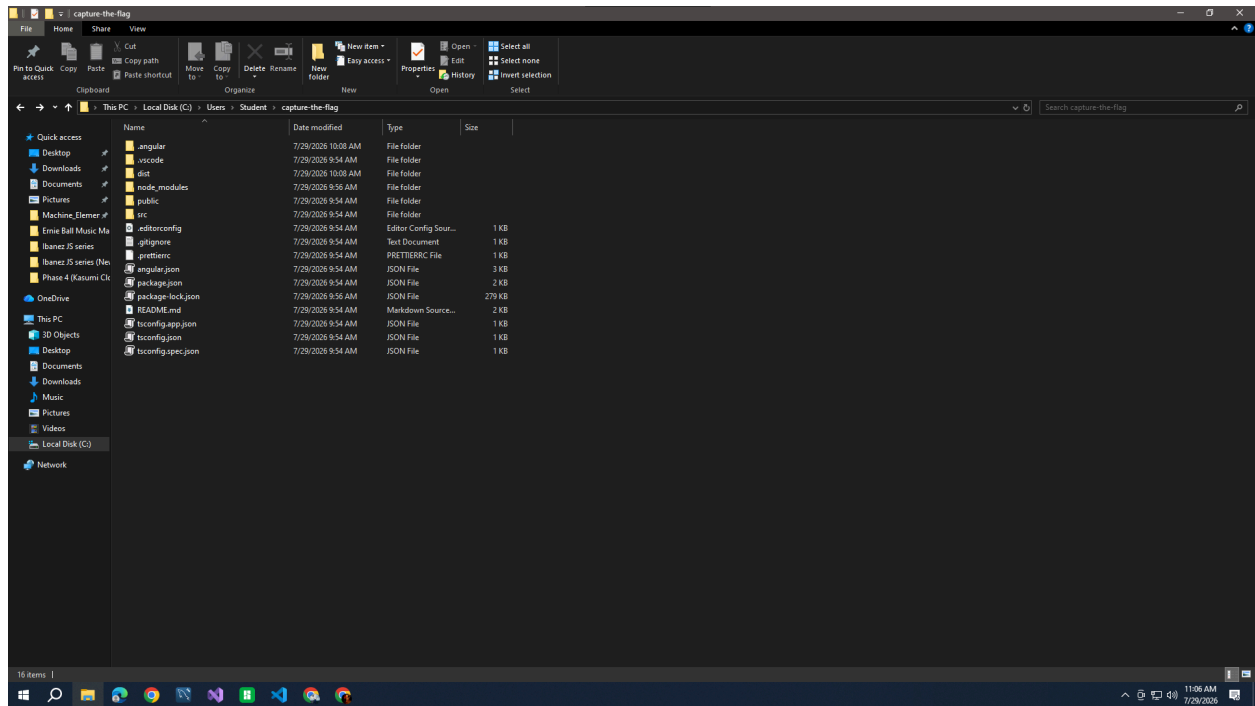


Flag 7 – Find the Entry Point

Objective: Figure out where the default Angular page comes from.

Screenshot:

- The location of the file where the default page is from



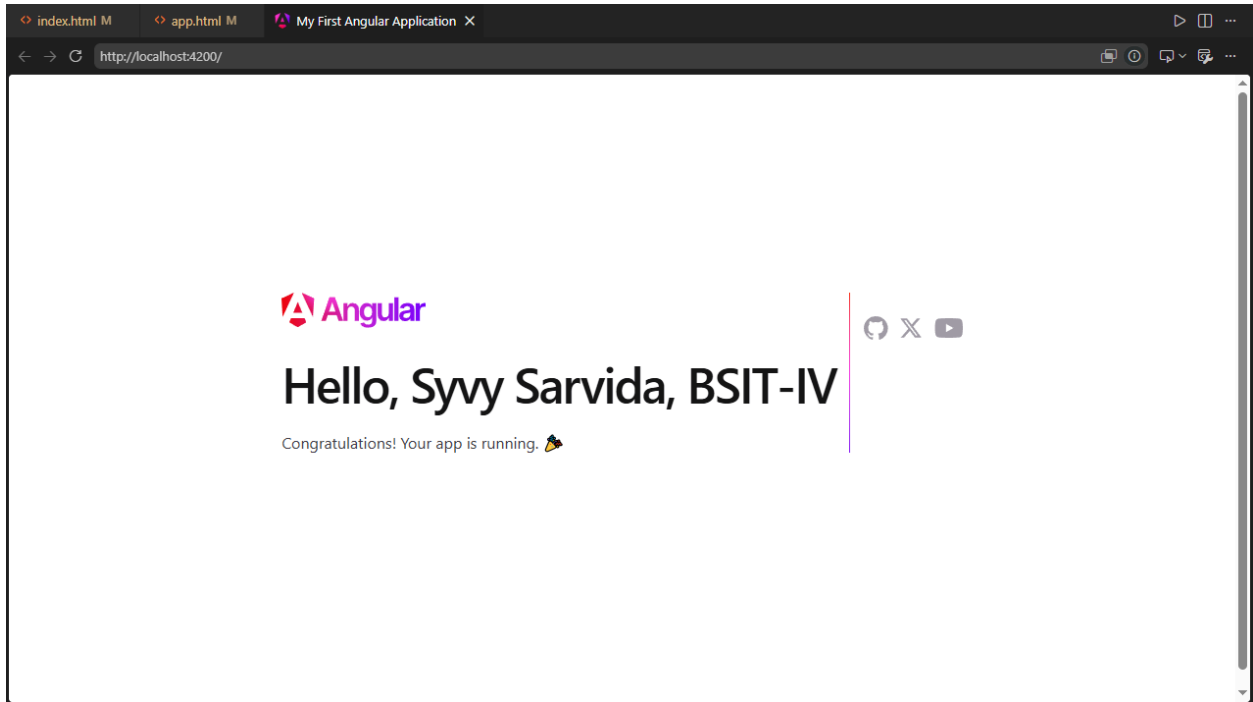
Flag 8 – Claim Your App

Replace the default content with:

- Your full name
- Your course & section
- "My First Angular Application"
- Today's date

Screenshot:

- Your updated content of the code
- The updated browser UI



Flag 9 – Personalize It

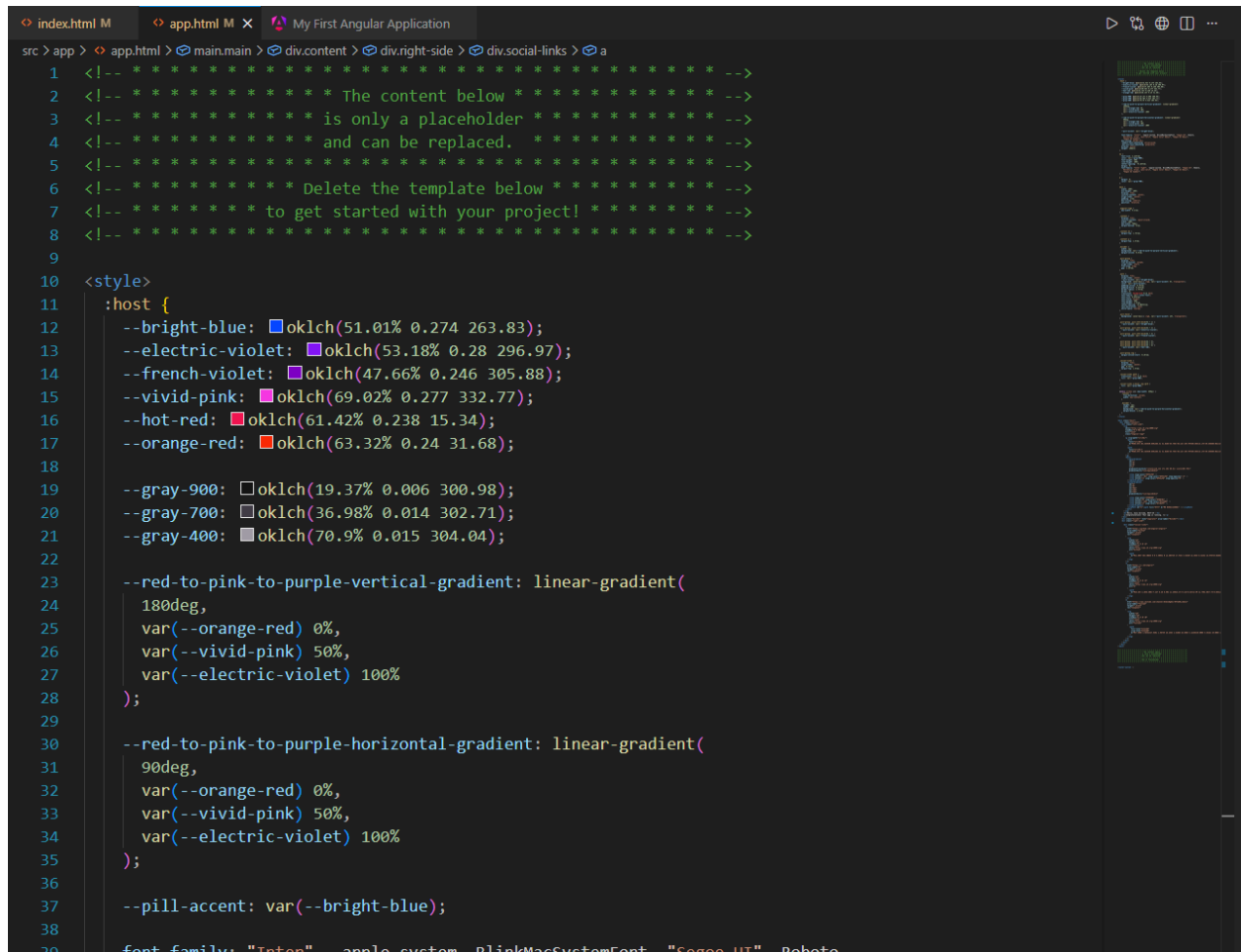
Without following a step-by-step tutorial, make **at least three** visual changes by researching how.

Examples:

- Change the background colour.
- Change the text colour.
- Centre the content.
- Increase font size.
- Add a border or rounded card.
- Change the font.
- Add an emoji or icon.

Screenshot:

- The code changed for personalizing
- The updated browser UI



```
src > app > app.html > main.main > div.content > div.right-side > div.social-links > a
1 <!-- ***** -->
2 <!-- ***** The content below ***** -->
3 <!-- ***** is only a placeholder ***** -->
4 <!-- ***** and can be replaced. ***** -->
5 <!-- ***** -->
6 <!-- ***** Delete the template below ***** -->
7 <!-- ***** to get started with your project! ***** -->
8 <!-- ***** -->
9
10 <style>
11 :host {
12   --bright-blue: oklch(51.01% 0.274 263.83);
13   --electric-violet: oklch(53.18% 0.28 296.97);
14   --french-violet: oklch(47.66% 0.246 305.88);
15   --vivid-pink: oklch(69.02% 0.277 332.77);
16   --hot-red: oklch(61.42% 0.238 15.34);
17   --orange-red: oklch(63.32% 0.24 31.68);
18
19   --gray-900: oklch(19.37% 0.006 300.98);
20   --gray-700: oklch(36.98% 0.014 302.71);
21   --gray-400: oklch(70.9% 0.015 304.04);
22
23   --red-to-pink-to-purple-vertical-gradient: linear-gradient(
24     180deg,
25     var(--orange-red) 0%,
26     var(--vivid-pink) 50%,
27     var(--electric-violet) 100%
28   );
29
30   --red-to-pink-to-purple-horizontal-gradient: linear-gradient(
31     90deg,
32     var(--orange-red) 0%,
33     var(--vivid-pink) 50%,
34     var(--electric-violet) 100%
35   );
36
37   --pill-accent: var(--bright-blue);
38
39   font-family: "Inter", apple-system, BlinkMacSystemFont, "Segoe UI", Roboto
```

Flag 10 – Add an Image

Research how to display an image in Angular.

It can be:

- Your favourite programming logo.
- Your favourite animal.
- Your own avatar.
- Any royalty-free image.

Screenshot:

- The code changed when adding an image
- The updated browser UI



Hello, Syvy Sarvida, BSIT-IV

Congratulations! Your app is running. 🎉

 Profile Avatar

